

Numerical Analysis

Cheatsheet — Mathematics 101

Errors & Approximations

Error Types

- Absolute: $E_a = |x - \tilde{x}|$
- Relative: $E_r = \frac{|x - \tilde{x}|}{|x|}$

Error Sources

- Round-off: finite precision
- Truncation: approximating infinite processes
- Input: measurement errors

Floating-Point $x = \pm m \times \beta^e$

Machine epsilon: $\epsilon_m \approx 2.2 \times 10^{-16}$ (double)

Catastrophic Cancellation Subtracting nearly equal numbers loses precision.

Fix: Rationalize or reformulate

Root Finding

Bisection Method Given $f(a) \cdot f(b) < 0$:

1. $c = \frac{a+b}{2}$
2. If $f(a) \cdot f(c) < 0$: $b = c$
3. Else: $a = c$
4. Repeat

Error: $|r - c_n| \leq \frac{b-a}{2^{n+1}}$

Convergence: **Linear** $O(1/2^n)$

Newton-Raphson

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Convergence: **Quadratic**

Digits double each iteration

Fails if: $f'(x_n) = 0$, bad initial guess

Secant Method

$$x_{n+1} = x_n - f(x_n) \frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})}$$

No derivative needed

Convergence: $O(\phi) \approx O(1.618)$

Fixed-Point Iteration Rewrite $f(x) = 0$

as $x = g(x)$

$$x_{n+1} = g(x_n)$$

Converges if $|g'(r)| < 1$

Interpolation

Lagrange Polynomial

$$P_n(x) = \sum_{i=0}^n y_i L_i(x)$$

where

$$L_i(x) = \prod_{j \neq i} \frac{x - x_j}{x_i - x_j}$$

Newton's Divided Diff

$$P_n(x) = \sum_{k=0}^n f[x_0, \dots, x_k] \prod_{j=0}^{k-1} (x - x_j)$$

Divided Differences:

$$\begin{aligned} f[x_i] &= y_i \\ f[x_i, x_j] &= \frac{f[x_j] - f[x_i]}{x_j - x_i} \end{aligned}$$

Interpolation Error

$$f(x) - P_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} \prod_{i=0}^n (x - x_i)$$

Runge's Phenomenon: High-degree + equally spaced = oscillations

Numerical Differentiation

Finite Differences Forward:

$$f'(x) \approx \frac{f(x+h) - f(x)}{h} + O(h)$$

Backward:

$$f'(x) \approx \frac{f(x) - f(x-h)}{h} + O(h)$$

Central:

$$f'(x) \approx \frac{f(x+h) - f(x-h)}{2h} + O(h^2)$$

Second Derivative

$$f''(x) \approx \frac{f(x+h) - 2f(x) + f(x-h)}{h^2}$$

Warning: Differentiation amplifies errors!

Numerical Integration

Trapezoidal Rule

$$\int_a^b f \, dx \approx \frac{h}{2} [f(a) + f(b)]$$

Composite:

$$\frac{h}{2} \left[f_0 + 2 \sum_{i=1}^{n-1} f_i + f_n \right]$$

Error: $O(h^2)$

Simpson's Rule

$$\int_a^b f \, dx \approx \frac{h}{3} [f_0 + 4f_1 + f_2]$$

Composite (n even):

$$\frac{h}{3} \left[f_0 + 4 \sum_{\text{odd}} + 2 \sum_{\text{even}} + f_n \right]$$

Error: $O(h^4)$

Comparison

Trapezoidal	$O(h^2)$
Simpson's	$O(h^4)$

Linear Systems

Gaussian Elimination

1. Forward elimination $\rightarrow$ upper triangular
2. Back substitution

Multiplier: $m_{ik} = \frac{a_{ik}}{a_{kk}}$

Partial Pivoting: Swap rows to maximize $|a_{kk}|$

Back Substitution

$$x_i = \frac{1}{a_{ii}} \left(b_i - \sum_{j>i} a_{ij} x_j \right)$$

LU Decomposition $A = LU$ (L lower, U upper)

1. Solve $Ly = b$ (forward)
2. Solve $Ux = y$ (backward)

Iterative Methods Jacobi:

$$x_i^{(k+1)} = \frac{1}{a_{ii}} \left(b_i - \sum_{j \neq i} a_{ij} x_j^{(k)} \right)$$

Gauss-Seidel: Use updated values immediately

Converges if: Diagonally dominant $|a_{ii}| > \sum_{j \neq i} |a_{ij}|$

Numerical ODEs

IVP $\frac{dy}{dt} = f(t, y)$, $y(t_0) = y_0$

Euler's Method

$$y_{n+1} = y_n + h \cdot f(t_n, y_n)$$

Error: $O(h)$ (first-order)

Runge-Kutta 4 (RK4)

$$k_1 = hf(t_n, y_n)$$

$$k_2 = hf(t_n + \frac{h}{2}, y_n + \frac{k_1}{2})$$

$$k_3 = hf(t_n + \frac{h}{2}, y_n + \frac{k_2}{2})$$

$$k_4 = hf(t_n + h, y_n + k_3)$$

$$y_{n+1} = y_n + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

Error: $O(h^4)$ (fourth-order)

Method Comparison

Euler	$O(h)$, 1 eval
RK2	$O(h^2)$, 2 evals
RK4	$O(h^4)$, 4 evals